

DAY-3

```
jupyter day-3 Last Checkpoint: 14 minutes ago

File Edit View Run Kernel Settings Help Trusted

JupyterLab Python [conda env:base] Anaconda Toolbox

[1]: !pip install pandas

Requirement already satisfied: pandas in c:\users\lenovo\anaconda3\lib\site-packages (2.2.2)
Requirement already satisfied: numpy>=1.26.0 in c:\users\lenovo\anaconda3\lib\site-packages (from pandas) (1.26.4)
Requirement already satisfied: python-dateutil>=2.8.2 in c:\users\lenovo\anaconda3\lib\site-packages (from pandas) (2.9.0.post0)
Requirement already satisfied: pytz>=2020.1 in c:\users\lenovo\anaconda3\lib\site-packages (from pandas) (2024.1)
Requirement already satisfied: tzdata>=2022.7 in c:\users\lenovo\anaconda3\lib\site-packages (from pandas) (2023.3)
Requirement already satisfied: six>=1.5 in c:\users\lenovo\anaconda3\lib\site-packages (from python-dateutil>=2.8.2->pandas) (1.16.0)

[3]: import pandas as pd
a=pd.Series([1,2,3,4,5])
print(a)

0    1
1    2
2    3
3    4
4    5
dtype: int64

[4]: print(a.dtype)

int64

[5]: print(a.index)

RangeIndex(start=0, stop=5, step=1)

[6]: a=pd.Series([True,False,True])
print(a)

0     True
1    False
2     True
dtype: bool
```

```
localhost:8889/notebooks/datascience%20day-3.ipynb Verify it's you ...

jupyter day-3 Last Checkpoint: 14 minutes ago

File Edit View Run Kernel Settings Help Trusted

JupyterLab Python [conda env:base] Anaconda Toolbox

[7]: print(a.dtype)

bool

[8]: print(a.index)

RangeIndex(start=0, stop=3, step=1)

[11]: arr=pd.Series([31,25,15,45,81,69,88,],index=['A','B','C','D','E','F','G','H'])
print(arr)

A    31
B    25
C    15
D    45
E    81
F    69
G    88
dtype: int64

[12]: print(arr['E']) #displays E th value

81

[13]: print(arr+5) #adding 5 to all the values

A    36
B    30
C    20
D    50
E    86
F    74
G    93
H    93
dtype: int64

[14]: print(arr.iloc[1])

25
```

localhost:8889/notebooks/datascience%2Fday-3.ipynb

Jupyter day-3 Last Checkpoint: 14 minutes ago

File Edit View Run Kernel Settings Help

JupyterLab Python [conda env:base] Anaconda Toolbox

```
[17]: print(arr.loc['A'])
31

[18]: #STATISTICAL FUNCTIONS
print(arr.sum())
354

[19]: print(arr.mean())
50.57142857142857

[20]: print(arr.median())
45.0

[21]: print(arr.mode())
0    15
1    25
2    31
3    45
4    69
5    81
6    88
dtype: int64

[22]: print(arr.min())
15

[23]: print(arr.max())
88
```

Jupyter day-3 Last Checkpoint: 14 minutes ago

File Edit View Run Kernel Settings Help

JupyterLab Python [conda env:base] Anaconda Toolbox

```
[24]: print(arr.std())
28.86668866159945

[25]: print(arr[arr>30])
A    31
D    45
E    81
FG   69
H    88
dtype: int64

[26]: print(arr.value_counts())
31    1
25    1
15    1
45    1
81    1
69    1
88    1
Name: count, dtype: int64

[27]: name=pd.Series(['pen','pencil','book','note','paper'])
print(name.str.upper())
0      PEN
1  PENCIL
2   BOOK
3   NOTE
4   PAPER
dtype: object

[28]: print(name.str.len())
0      3
1     6
2     4
3     4
4     5
```



```
A    31
D    45
E    81
FG   69
H    88
dtype: int64
```

```
[26]: print(arr.value_counts())
```

```
31    1
25    1
15    1
45    1
81    1
69    1
88    1
Name: count, dtype: int64
```

```
[27]: name=pd.Series(['pen','pencil','book','note','paper'])
      print(name.str.upper())
```

```
0      PEN
1  PENCIL
2     BOOK
3     NOTE
4     PAPER
dtype: object
```

```
[28]: print(name.str.len())
```

```
0     3
1     6
2     4
3     4
4     5
dtype: int64
```